

Dr. Doc:

A 12-year-old boy with severe refractory allergic asthma is prescribed a monoclonal antibody that selectively binds to and neutralizes Interleukin-5 (IL-5). Which of the following peripheral blood cell types will decrease most significantly as a direct result of this therapy?

- A. Neutrophils
- B. Eosinophils
- C. Basophils
- D. B-lymphocytes

Correct Answer: B. Eosinophils

Explanation: IL-5 is the primary cytokine responsible for the differentiation, recruitment, activation, and survival of eosinophils. Blocking IL-5 (e.g., with Mepolizumab) markedly depletes eosinophil counts.

During the initial sensitization phase of atopic asthma, naive B cells undergo class-switch recombination to produce antigen-specific IgE antibodies. Which cytokine secreted by Th2 helper cells is primarily responsible for driving this class switch?

- A. Interferon-gamma
- B. Interleukin-2
- C. Interleukin-4
- D. Tumor Necrosis Factor-alpha

Correct Answer: C. Interleukin-4 (IL-4)

Explanation: IL-4 (along with IL-13) stimulates B cells to undergo heavy-chain class switching from IgM to IgE. IFN- γ is produced by Th1 cells and inhibits Th2 responses.

Which congenital anomaly is characterized by a discrete area of lung tissue that lacks any connection to the airway system and has an abnormal blood supply arising from the aorta?

- A) Pulmonary hypoplasia
- B) Foregut cysts
- C) Pulmonary sequestration
- D) Tracheoesophageal fistula

Correct Answer: C

- Option A: Pulmonary hypoplasia is a congenital defect in lung development leading to decreased lung volume, but it does not feature an isolated tissue mass with aortic blood supply.
- Option B: Foregut cysts are congenital abnormalities of embryological development but remain separate from sequestration pathology.
- Option C: Pulmonary sequestration is defined as a discrete area of lung tissue that lacks any connection to the airway system and has an abnormal blood supply arising from the aorta or its branches.
- Option D: Tracheoesophageal fistula is an abnormal connection between the trachea and esophagus, often accompanied by esophageal atresia.

What is the most common anatomical pattern of Tracheoesophageal Fistula (TEF) and Esophageal Atresia?

- A) Double fistula with both proximal and distal esophagus communicating with the trachea
- B) Blind proximal esophagus with a fistula from the distal esophagus to the trachea
- C) Isolated esophageal atresia without any fistula to the trachea
- D) Fistula from the proximal esophagus to the trachea with a normal distal esophagus

Correct Answer: B

Which form of lung collapse stems directly from complete obstruction of an airway?

- A) Compression atelectasis

- B) Contraction atelectasis
- C) Resorption atelectasis
- D) Relaxation atelectasis

Correct Answer: C

- Option A: Compression atelectasis is typically caused by the accumulation of fluid, blood, or air within the pleural cavity, compressing the lung from the outside.
- Option B: Contraction atelectasis is caused by focal or generalized pulmonary or pleural fibrosis that physically prevents full expansion.
- Option C: Resorption atelectasis stems from complete obstruction of an airway, preventing air from reaching distal airways, which leads to the absorption of trapped air and subsequent alveolar collapse.
- Option D: Relaxation atelectasis is another term for compression atelectasis.

Which clinical definition corresponds to chronic bronchitis?

- A) Permanent dilation of bronchi and bronchioles with destruction of elastic tissue
- B) Persistent cough with sputum production for at least 3 months in at least 2 consecutive years
- C) Irreversible enlargement of airspaces distal to the terminal bronchiole
- D) Recurrent episodes of wheezing and dyspnea triggered by type I hypersensitivity

Correct Answer: B

- Option A: Permanent bronchial dilation with tissue destruction describes bronchiectasis.
- Option B: Chronic bronchitis is clinically defined as persistent cough with sputum production for at least 3 months in at least 2 consecutive years, in the absence of any other cause.
- Option C: Irreversible enlargement of the airspaces distal to the terminal bronchiole describes emphysema.
- Option D: Recurrent wheezing episodes triggered by hypersensitivity describes atopic asthma.

What is the normal value of the Reid index in a healthy individual?

- A) 0.2
- B) 0.4
- C) 0.6
- D) 0.8

Correct Answer: B

Which leukocyte is locally recruited and activated by Interleukin-5 (IL-5) in the pathogenesis of asthma?

- A) Eosinophils
- B) Neutrophils
- C) Basophils
- D) Alveolar macrophages

Correct Answer: A

Which eosinophil protein constitutes the Charcot-Leyden crystals found in asthmatic sputum?

- A) Galectin-10
- B) Major basic protein
- C) Histamine
- D) Interleukin-4

Correct Answer: A

Which morphological type of emphysema is most strongly associated with a congenital alpha1-antitrypsin (alpha1-AT) deficiency?

- A) Centriacinar emphysema
- B) Panacinar emphysema

C) Paraseptal emphysema

D) Irregular emphysema

Correct Answer: B

- Option A: Centriacinar emphysema is heavily associated with cigarette smoking and spares the distal acinus.
- Option B: Panacinar emphysema, where acini are uniformly enlarged and most severe at the bases, is associated with alpha1-antitrypsin deficiency.
- Option C: Paraseptal emphysema involves the distal part of the acinus and is associated with localized scarring, not alpha1-AT deficiency.
- Option D: Irregular emphysema is associated with scars and is clinically insignificant.

In which region of the lungs does centriacinar emphysema predominantly manifest?

A) Lower lobes

B) Upper lobes

C) Anterior margins

D) Diffusely across both lungs

Correct Answer: B

Which non-bacterial organism is a common cause of atypical pneumonia?

A) Streptococcus pneumoniae

B) Mycoplasma pneumoniae

C) Klebsiella pneumoniae

D) Pseudomonas aeruginosa

Correct Answer: B

- Option A: *Streptococcus pneumoniae* is the classic bacterium responsible for typical lobar pneumonia.
- Option B: Causes of atypical pneumonia include *Mycoplasma*, viruses (such as RSV and influenza), and *Chlamydia*.
- Option C: *Klebsiella pneumoniae* causes typical bacterial pneumonia with necrosis.
- Option D: *Pseudomonas aeruginosa* is a bacterium associated with nosocomial typical pneumonia.

Which predisposing clinical condition is most strongly associated with the development of a lung abscess?

- A) Active cigarette smoking
- B) Chronic alcoholism
- C) Primary ciliary dyskinesia
- D) Severe atopic asthma

Correct Answer: B

- Option A: Smoking predisposes to chronic bronchitis and emphysema, but is not the primary direct risk factor for lung abscess.
- Option B: Alcoholism is the most common condition predisposing to lung abscess, due to impaired consciousness and high risk of gastric content aspiration.
- Option C: Ciliary dyskinesia is associated with bronchiectasis, not primarily isolated lung abscess.
- Option D: Asthma causes episodic bronchoconstriction and is not a primary predisposing factor for abscesses.

In which anatomical site is a single, aspiration-induced lung abscess most commonly located?

- A) Left lung
- B) Right lung
- C) Bilateral upper lobes
- D) Bilateral bases

Correct Answer: B

Which histological feature is the defining pathognomonic hallmark of acute respiratory distress syndrome (ARDS)?

- A) Noncaseating granulomas
- B) Curschmann spirals
- C) Hyaline membranes
- D) Fibrinosuppurative consolidation

Correct Answer: C

- Option A: Noncaseating granulomas are the hallmark of sarcoidosis.
- Option B: Curschmann spirals are diagnostic sputum findings in asthma.
- Option C: The histologic manifestation of ALI/ARDS is diffuse alveolar damage (DAD), with hyaline membranes being a characteristic feature.
- Option D: Fibrinosuppurative consolidation is the histologic pattern of typical bacterial lobar pneumonia.

What is the classic clinical description for a patient with severe emphysema who overventilates to remain well-oxygenated?

- A) Blue bloater
- B) Pink puffer
- C) Cobble breather
- D) Silent lunger

Correct Answer: B

- Option A: Blue bloaters refers to patients with chronic bronchitis who present with hypoxemia and cyanosis.

- Option B: Classically, the patient with severe emphysema is barrel-chested, dyspneic, sits forward, breathes through pursed lips, and may overventilate to remain well-oxygenated (Pink Puffer).
- Option C: Cobblestoning is a gross pleural finding in IPF, not a clinical term for patients.
- Option D: Silent lunger is not a medical term.

Which of the following factors is specifically identified as interfering with the action of alveolar macrophages?

- A) Viral respiratory infections
- B) Loss of the cough reflex
- C) Alcohol consumption
- D) Complete airway obstruction

Correct Answer: C

- Option A: Viral respiratory infections cause injury to the mucociliary apparatus, not direct macrophage interference.
- Option B: Loss of the cough reflex is caused by coma, anesthesia, or drugs, and impairs pulmonary clearance but not by macrophage inhibition.
- Option C: The sources state that the action of alveolar macrophages can be impaired by interference from alcohol, tobacco, or anoxia.
- Option D: Complete airway obstruction causes resorption atelectasis, which is separate from cellular macrophage interference.

During which stage of typical lobar pneumonia does the affected lobe grossly appear red, firm, airless, and of liver-like consistency?

- A) Congestion
- B) Red hepatization
- C) Gray hepatization

D) Resolution

Correct Answer: B

- Option A: In the congestion stage, the lung is heavy, red, and boggy, but remains soft, and has not yet become firm or liver-like.
- Option B: Red hepatization is marked by a massive confluent exudation with red cells, neutrophils, and fibrin filling alveolar spaces; grossly, the lobe appears red, firm, airless, with a liver-like consistency.
- Option C: Gray hepatization is marked by progressive disintegration of red cells and persistence of a grayish-brown fibrinosuppurative exudate.
- Option D: Resolution is the final stage where the exudate is digested and resorbed.

Which direction does the mediastinum shift in a patient with compression atelectasis?

- A) Toward the atelectatic (affected) lung
- B) Away from the affected lung
- C) Downward toward the abdominal cavity
- D) No shift occurs in this form of collapse

Correct Answer: B

- Option A: The mediastinum shifts toward the affected lung in resorption atelectasis.
- Option B: In compression atelectasis, pressure from fluid, blood, or air in the pleural space pushes the lung and causes the mediastinum to shift away from the affected lung.
- Option C: Downward displacement is diaphragmatic, not mediastinal.
- Option D: A mediastinal shift is a crucial clinical feature distinguishing resorption and compression atelectasis.

Which set of pulmonary function test results is characteristic of an obstructive lung disease?

- A) Increased total lung capacity and an FEV1/FVC ratio greater than 0.85

- B) Decreased maximal airflow rates during forced expiration, with an FEV1/FVC ratio less than 0.7
- C) Proportionate decreases in total lung capacity and FEV1, resulting in a normal FEV1/FVC ratio
- D) Reduced FVC with normal or elevated expiratory flow rate

Correct Answer: B

- Option A: Obstructive diseases characteristically feature a decreased FEV1/FVC ratio.
- Option B: Obstructive diseases are characterized by a decrease in maximal airflow rates during forced expiration (decreased FEV1/FVC ratio); a ratio of less than 0.7 generally indicates obstruction.
- Option C: Proportionate decreases in TLC and FEV1 with a normal ratio are characteristic of restrictive lung diseases.
- Option D: Reduced FVC with normal flow rate is typical of restrictive, not obstructive, lung disease.

Distal acinar (paraseptal) emphysema is characteristically associated with which of the following clinical events?

- A) Severe alpha1-antitrypsin deficiency presenting in childhood
- B) Spontaneous pneumothorax in young adults
- C) Weight loss so severe that it suggests an occult cancer
- D) Development of cor pulmonale and right-sided congestive heart failure

Correct Answer: B

- Option A: Panacinar emphysema, not distal acinar, is associated with alpha1-antitrypsin deficiency.
- Option B: Distal acinar (paraseptal) emphysema underlies many cases of spontaneous pneumothorax.
- Option C: Severe weight loss mimicking occult cancer is a feature of advanced centriacinar or panacinar emphysema.
- Option D: Right-sided failure is a terminal complication of generalized, severe emphysema, whereas distal acinar is often localized and silent until rupture.

How does tobacco smoke exposure contribute to the development of a functional alpha1-antitrypsin deficiency in emphysema?

- A) By causing genetic mutations in the alpha1-AT gene in liver cells
- B) By producing oxidants that chemically inactivate protective antiproteases
- C) By blocking the secretion of antiproteases from macrophages
- D) By directly degrading elastase before it can reach the lungs

Correct Answer: B

Which clinical syndrome features a primary defect in ciliary clearance leading to a triad of bronchiectasis, sinusitis, and situs inversus?

- A) Kartagener syndrome
- B) Sarcoidosis
- C) Idiopathic pulmonary fibrosis
- D) Status asthmaticus

Correct Answer: A

What is the characteristic gross appearance of the pleural surfaces in a patient with advanced idiopathic pulmonary fibrosis (IPF)?

- A) Smooth and glistening with focal subpleural blebs
- B) Cobblestoned due to retraction of scars along the interlobular septa
- C) Covered in a thick, yellowish-green purulent exudate
- D) Diffusely hyperinflated and voluminous, meeting in the midline

Correct Answer: B

Which inorganic dust inhalation is associated with nodular fibrosis in the lungs?

- A) Asbestos fibers

- B) Silica dust
- C) Coal dust
- D) Beryllium particles

Correct Answer: B

- Option A: Asbestosis causes diffuse interstitial fibrosis and pleural plaques, not localized nodular fibrosis.
- Option B: Silicosis is characterized by nodular fibrosis resulting from the inhalation of silica dust.
- Option C: Coal worker's pneumoconiosis leads to coal macules/nodules but is distinct from classical nodular silicosis.
- Option D: Beryllium causes a sarcoid-like granulomatous reaction, not classical nodular fibrosis.

Which diagnostic morphological finding is characteristically observed in sarcoidosis, regardless of the organ involved?

- A) Caseating epithelioid granuloma
- B) Noncaseating epithelioid granuloma
- C) Fibrinosuppurative alveolar consolidation
- D) Interstitial honeycomb fibrosis

Correct Answer: B

- Option A: Caseating granulomas are classic for tuberculosis, whereas sarcoidosis granulomas are characteristically noncaseating.
- Option B: Sarcoidosis is morphologically characterized by noncaseating epithelioid granulomas.
- Option C: Fibrinosuppurative consolidation is the hallmark of typical bacterial lobar pneumonia.
- Option D: Honeycomb fibrosis is the microscopic hallmark of IPF.

What are Schaumann bodies, which are commonly found in the granulomas of sarcoidosis?

- A) Star-shaped inclusion bodies within giant cells
- B) Laminated concretions composed of calcium and proteins
- C) Crystalline remnants of degenerated eosinophil granules
- D) Mucus plugs extruded from salivary gland ducts

Correct Answer: B

- Option A: Star-shaped inclusions are Asteroid bodies, which are also found in sarcoidosis giant cells.
- Option B: Schaumann bodies are laminated concretions composed of calcium and proteins.
- Option C: Eosinophil protein remnants are Charcot-Leyden crystals.
- Option D: Sputum mucus plugs in asthma are Curschmann spirals.

Which anatomical pattern of bacterial pneumonia is morphologically characterized by a patchy consolidation of the lung tissue?

- A) Lobar pneumonia
- B) Bronchopneumonia
- C) Interstitial pneumonia
- D) Atypical pneumonia

Correct Answer: B

Which organism is particularly noted in the source for causing lung tissue destruction and necrosis leading to abscess formation during pneumonia?

- A) *Mycoplasma pneumoniae*
- B) Type 3 pneumococci or *Klebsiella*
- C) Rhinovirus
- D) *Streptococcus pneumoniae* type 1

Correct Answer: B

What are Asteroid bodies, which can be found in the granulomas of a patient with sarcoidosis?

- A) Laminated calcium-protein concretions
- B) Star-shaped inclusion bodies within multinucleated giant cells
- C) Lipofuscin deposits in alveolar epithelial cells
- D) Crystallized immunoglobulin complexes in macrophages

Correct Answer: B

A patient presents with sudden onset of high-grade fever, chills, and productive cough with 'rust-colored' sputum. Which organism is classically responsible for this presentation?

- A) *Mycoplasma pneumoniae*
- B) *Streptococcus pneumoniae*
- C) *Klebsiella pneumoniae*
- D) *Pseudomonas aeruginosa*

Correct Answer: B

Which group of organisms is most commonly isolated in nosocomial pneumonia?

- A) *Mycoplasma pneumoniae* and respiratory syncytial virus
- B) *Streptococcus pneumoniae* and *Hemophilus influenzae*
- C) *S. aureus*, Enterobacteriaceae, and *Pseudomonas*
- D) *Bacteroides* and *Fusobacterium*

Correct Answer: C

Which diagnostic marker or feature is highly characteristic of atopic asthma?

- A) Normal serum IgE levels and negative skin tests

- B) Association with a negative family history
- C) Positive skin test to allergen and history of allergic rhinitis, urticaria, or eczema
- D) Initial presentation primarily in late adulthood

Correct Answer: C

A 45-year-old bedridden man who recently underwent abdominal surgery for severe ascites develops sudden respiratory distress. Physical exam reveals decreased breath sounds at the bases of both lungs. A chest X-ray shows bibasilar areas of collapse, and the mediastinum is noted to be shifted slightly away from the affected areas. What is the most likely pathological mechanism?

- A) Complete airway obstruction by a mucus plug
- B) Accumulation of fluid or elevated diaphragm position
- C) Localized pulmonary or pleural fibrosis
- D) Exaggerated TH2 response and bronchospasm

Correct Answer: B

- Option A: Obstruction of an airway causes resorption atelectasis, which shifts the mediastinum TOWARD, not away from, the affected lung.
- Option B: Compression atelectasis (passive or relaxation atelectasis) is associated with diaphragm elevation (common in bedridden patients and those with ascites, especially postoperatively) and shifts the mediastinum AWAY from the affected lung.
- Option C: Contraction atelectasis occurs when focal or generalized fibrosis prevents full lung expansion and is typically chronic, not presenting with acute postoperative diaphragmatic shift.
- Option D: Asthma causes reversible airway obstruction but not acute structural collapse with mediastinal shift away.

A 28-year-old woman with a history of recurrent respiratory infections and chronic sinus issues presents with a persistent, severe cough that produces large volumes of foul-smelling, purulent sputum. Her fingers show prominent clubbing. A chest CT shows marked dilation of the bronchi in both lower lobes. What is the most likely diagnosis?

- A) Centriacinar emphysema

B) Chronic bronchitis

C) Bronchiectasis

D) Idiopathic pulmonary fibrosis

Correct Answer: C

A 68-year-old retired sandblaster presents with progressive dyspnea and a dry cough. Pulmonary function tests show a restrictive pattern (normal FEV1/FVC ratio and reduced total lung capacity). A chest X-ray shows multiple small, hard nodules in the upper lobes. What is the most likely diagnosis?

A) Asbestosis

B) Silicosis

C) Coal worker's pneumoconiosis

D) Sarcoidosis

Correct Answer: B

A 35-year-old African American woman presents with a dry cough, shortness of breath, and red, tender skin nodules on her shins. A chest X-ray shows prominent bilateral hilar lymphadenopathy. A biopsy of a lymph node reveals noncaseating epithelioid granulomas with asteroid bodies. What is the most likely diagnosis?

A) Pulmonary tuberculosis

B) Sarcoidosis

C) Silicosis

D) Atypical pneumonia

Correct Answer: B

A 72-year-old man with severe dysphagia secondary to a recent stroke is admitted to the hospital after developing a high fever, tachypnea, and a productive cough. He has a history of severe gastroesophageal reflux disease (GERD). On auscultation, coarse crackles are heard in the right lower lobe. If he survives the acute episode, which complication is most commonly associated with this type of pneumonia?

- A) Spontaneous pneumothorax
- B) Honeycomb lung and diffuse fibrosis
- C) Lung abscess formation
- D) Bilateral hilar lymphadenopathy

Correct Answer: C

- Option A: Pneumothorax is a complication of distal acinar emphysema, not primarily aspiration pneumonia.
- Option B: Honeycomb lung is the end-stage of chronic interstitial diseases like IPF.
- Option C: Aspiration pneumonia occurs in debilitated patients or those who aspirate gastric contents. It is often necrotizing, and in those who survive, lung abscess formation is a common complication.
- Option D: Hilar lymphadenopathy is classic for sarcoidosis.

A 48-year-old chronic alcoholic is brought to the emergency department in a state of severe intoxication. Three weeks later, he develops a high fever, a cough productive of foul-smelling sputum, and finger clubbing. A chest X-ray reveals a single 3.5 cm cavity with an air-fluid level in the right lower lobe. What is the most likely diagnosis?

- A) Typical pneumococcal lobar pneumonia
- B) Acute severe asthma
- C) Primary lung abscess
- D) Extralobar pulmonary sequestration

Correct Answer: C

A 28-year-old man who is a habitual smoker presents with episodic wheezing and a cough that is dry. His serum IgE levels are normal, and skin prick tests to common environmental allergens are negative. He reports that his symptoms are almost always preceded by a common cold. What is the most likely diagnosis?

- A) Atopic asthma
- B) Non-atopic asthma

C) Centriacinar emphysema

D) Chronic bronchitis

Correct Answer: B

A 65-year-old man with a long-standing history of smoking presents with progressive dyspnea and significant weight loss. He has a barrel-chested appearance, sits in a hunched-over position, and breathes through pursed lips. His skin is well-oxygenated. Pulmonary function tests show an FEV1/FVC ratio of 0.5. What is the most likely diagnosis?

A) Idiopathic pulmonary fibrosis

B) Chronic bronchitis

C) Emphysema

D) Sarcoidosis

Correct Answer: C

- Option A: IPF is a restrictive disease with a normal FEV1/FVC ratio and is not associated with a barrel chest.
- Option B: Chronic bronchitis patients are classically 'blue bloaters' (cyanotic and overweight), not 'pink puffers' (thin, barrel-chested, well-oxygenated).
- Option C: The patient presents as a classic 'pink puffer' with severe emphysema (barrel-chested, dyspneic, sits forward, pursed-lip breathing, weight loss, and obstructive PFT with FEV1/FVC ratio < 0.7).
- Option D: Sarcoidosis is a restrictive disease characterized by granulomas, not a barrel chest and obstructive PFT.